LLM ONCOLOGY TRIAL LITERATURE REVIEW 2025, 2026

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July 27, 2026
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ABSTRACT

An author literature review was conducted with the purpose of gaining a better understanding of how recent Google Scholar articles effectively utilize large language models (LLMs). The review's plan was to utilize search terms "ai oncology", "ai oncology trials", and "ai pancreatic oncology trials" using dates since 2025 and 2026. Abstracts, figures, and conclusions were manually screened for LLM applications. For each of the study's seven sessions, twelve relevant papers were identified that met LLM objectives. A selection of key articles are included that detail A) Persistent publication of previously released LLMs, B) Lack of Phase 1/2/3 End-to-end LLM workflows, C) Many articles are based on the oncology trial patient enrollment stage, D) Large number of review articles have been published, and E) Broad speculation exists regarding how LLMs will be utilized for oncology in the future. In conclusion, these five observations indicate a lack of effective and scalable oncology trial workflows readily accessible with today's state-of-the art LLMs.

Keywords LLM Oncology Trial Literature Review; Artificial Intelligence; Pancreatic Cancer; Patient Enrollment

A) Persistent Publication of Previously Released LLMs

1. **Title:** "Artificial intelligence-based biomarkers for treatment decisions in oncology" 1/14/25 [1]
   Cons: The authors used GPT-4 to improve the readability and language of the manuscript.
   Results: OpenAI's successor to GPT-4 was GPT-4o - released in May 2024, 9 months prior to the publication date. GPT-4o generated text tokens approximately twice as fast as GPT-4, was natively multimodal, and half the cost as the author's model; making GPT-4o an improved alternative [2].
2. **Title:** "AI in critical care: A roadmap to the future" 9/23/25 [3]
   Cons: "During the writing process of the manuscript, some authors utilized GPT-4o (OpenAI) for linguistic purposes. All authors reviewed and edited the content as needed and take full responsibility for the content of the publication."
   Results: "Generative AI is expected to drive transformative changes across society and professional domains, including medicine and critical care." Paper was received 8/2/25, accepted on 8/21/25, and available online 9/23/25, which is over three months after OpenAI o3-pro with advanced reasoning and higher fidelity was released on 6/10/25 [4].
3. **Title:** "Discrepancies Between MDT Recommendations and AI-Generated Decisions in Gynecologic Oncology: A Retrospective Comparative Cohort Study" 1/30/26 [5]
   Cons: "Artificial intelligence (AI) in the form of large language models (LLMs) has emerged as an attractive means to support clinical decision-making [6]. The newer versions of LLMs, including ChatGPT-4o and ChatGPT-5, show significant advances in terms of text comprehension and coherence in answers when applied to medical queries [7,8]."
   Results: Standardized anonymized case summaries were entered into ChatGPT 5.0, which was instructed to follow current ESGO guidelines. "Overall concordance for FIGO staging was 77.0%, while treatment-related decisions demonstrated lower discordance, particularly in chemotherapy (8.2%) and targeted therapy (6.8%)." The article was received on 12/26/25, and published on 1/30/26, which came after the release of two newer and available OpenAI models: ChatGPT 5.1, 11/12/25, and ChatGPT 5.2, 12/11/25, [6, 7].
4. **Title:** "Generative AI and medical oncologists: Be curious, but cautious!" 10/2/25 [8]
   Cons: In this evaluation, ChatGPT-4 occasionally incorrectly recommended aromatase inhibitors for a subset of patients with ductal carcinoma in situ (DCIS), even though these patients are not eligible for that treatment.
   Results: As a result, the model often recommended curative treatments, whereas the guidelines would indicate a palliative approach in such cases. The article was received on 5/20/25, and accepted 10/2/25 with authors utilizing GPT-4; which is 17 months after the GPT-4o successor [2], and 4 months after the next gen o3-pro reasoning model [4].

B) Lack of Phase 1/2/3 End-to-End LLM Workflows

1. **Title:** "ChatGPT Versus DeepSeek: Assessing Artificial Intelligence Performance on Radiation Oncology Examination Questions" 10/24/25 [8]

Cons: "We prompted DeepSeek-R1 and several models of ChatGPT with 600 radiation oncology examination questions from national radiation oncology in-service multiple-choice examinations."
Results: "DeepSeek-R1 answered 84.0% of questions correctly, requiring 59 seconds per question." "ChatGPT o1 answered 89.0% of questions correctly, requiring 10 seconds per question." At February 2025 prices, DeepSeek-R1 cost was up to $1.56, compared with ChatGPT's $37.96. The paper was received 5/28/25, and published 10/24/25 - with the task featuring multiple-choice questions, and not an end-to-end LLM workflow.

2. **Title:** "Integrating Digital Health into Oncology: A Comprehensive Review" 1/07/26 [9]
Cons: "During the preparation of this manuscript, the authors used ChatGPT-5.2 to improve the structure, grammar, and language of the text. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication."
Results: The article was posted on 1/07/26 using a current ChatGPT-5.2 model, but the LLM was only utilized to improve basic paper formatting tasks, not for a comprehensive LLM workflow.
3. **Title:** "Challenges in the Clinical Application of Machine Learning for Pancreatic Cancer" 9/5/25 [10]
Cons: "The authors acknowledge the use of LLMs (e.g., ChatGPT, Gemini, Copilot) to assist in drafting portions of the manuscript, particularly in restructuring background information and refining language. The tool was not used for generating original scientific content, analyzing data, or drawing conclusions."
Results: Authors primarily focused on ML-driven tools, with LLMs utilized for "drafting portions of the manuscript", despite a scalable OpenAI o3 reasoning model release on 4/16/25, approximately three months prior to the paper being received on 7/7/25.
4. **Title:** "Exploring the Future of AI in Clinical Collaboration: A Study on Tumor Board Case Preparation" 4/13/26 [11]
Cons: "Recent research has begun to investigate the role that modern large-language models (LLMs) can play in MTBs [Multidisciplinary tumor boards], such as applications for surgical decision support, oncologic therapy, and precision oncology. However, to date, there have been no studies on clinicians' use or perception of applications of LLMs for supporting tumor board meetings."
Results: The tumor board case preparation consists of 30 authors from leading organizations including Stanford University, Northeastern University, and Microsoft. Tasks such as "Clinicians must extract key details from extensive records and evaluate treatment options." The use of "an off-the-shelf assistant (Copilot) and a task-specific multi-agent system (Healthcare Agent Orchestrator, HAO)" may have been better suited to the performant 2/5/26 Claude Code Opus 4.6 model for a larger End-to-end workflow [12]; provided that all data saw "additional techniques to ensure de-identification of the patient cases for use in the study, such as removing sensitive information and shifting the time."

C) Many Articles are Based on the Oncology Trial Patient Enrollment Stage

1. **Title:** "TrialMatchAI: an end-to-end AI-powered clinical trial recommendation system to streamline patient-to-trial matching" [13]
Cons: "To address this, future work should focus on developing robust flagging and monitoring mechanisms that allow clinicians to review AI-generated recommendations and report misclassifications, enabling continuous system improvement."
Results: A "real-world evaluation on biomarker-driven data from 52 metastatic cancer patients of the WIDE study demonstrated that 92% of patients had relevant trials successfully retrieved within the top 20 results."
2. **Title:** "Human-AI teaming to improve accuracy and efficiency of eligibility criteria prescreening for oncology trials: a randomized evaluation trial using retrospective electronic health records" [14]
Cons: Performance is limited in some domains due to automation bias. Although improvements are modest, this large randomized trial evaluating a human-AI framework for oncology prescreening shows that AI language models can approximate and augment human-driven prescreening to enhance identification of trial-eligible patients, potentially increasing enrollment.
Results: "Chart-level accuracy, the primary endpoint of Human+AI prescreening is noninferior and superior to Human-alone (76.5% vs. 71.1%). However, efficiency is unchanged with similar average time per chart review, the secondary endpoint, (37.4 vs. 37.8 min)."
3. **Title:** "307P AI-driven patient screening for clinical trials in pancreatic cancer: Integrating LLMs with tumor board meetings" [15]
Cons: Patient records presented at tumor board meetings between January 2018 and February 2023 were retrospectively reviewed based on clinical trials open for inclusion at that time ("candidate trials").
Results: "According to the gold standard, 175 were potentially eligible, while 166 had at least one unmet criterion. Mistral-7b-Instruct v0.3 demonstrated a high sensitivity (92.6%), outperforming GPT-4.5 (82.3%, p=0.01) and Claude 3.7 Sonnet (81.7%, p<.001); the specificities were not significantly different (respectively 92.9%; 91.0% and 90.1%)."
4. **Title:** "Clinical Trial Notifications Triggered by Artificial Intelligence–Detected Cancer Progression A Randomized Trial" 4/21/25 [16]
Cons: AI-triggered notifications "were no more likely to enroll in trials than were those in the control group."
Results: "The intervention had no significant impact on the trial enrollment rate (intervention, 2.20% [95% CI, 1.97%-2.46%]; control, 2.03% [95% CI, 1.72%-2.39%])."

D) Large Number of Review Articles Have Been Published

1. **Title:** "Concordance with CONSORT-AI guidelines in reporting of randomised controlled trials investigating artificial intelligence in oncology: a systematic review" 7/23/25 [17]
   Cons: Although the majority of CONSORT and CONSORT-AI items were well-reported, critical gaps related to reporting of methodology, reproducibility and harms persist.
   Results: This study included 57 RCTs of AI interventions in oncology that were primarily focused on screening (54%) or diagnosis (19%) and intended for clinician use (88%). Among all 57 RCTs, median concordance with CONSORT 2010 and CONSORT-AI 2020 was 82%.
2. **Title:** "Artificial intelligence and machine learning-driven advancements in gastrointestinal cancer: Paving the way for precision medicine" 1/7/26 [18]
   Cons: Recent trends have indicated an evolution in LLM implementation strategies. Information extraction and text classification have emerged the predominant NLP tasks in cancer research.
   Results: LLMs show promise for treatment decision support in oncology, though there are ongoing challenges regarding accuracy and reliability. Recent studies evaluating LLMs in suggesting treatment options found they generated a wider range of options compared to human experts, though their recommendations sometimes deviated from expert consensus.

E) Broad Speculation Exists Regarding how LLMs will be Utilized for Oncology in the Future

1. **Title:** "AI meets informed consent: a new era for clinical trial communication" 3/18/25 [19]
   Cons: "This study explores the potential of Large Language Models (LLMs), specifically GPT-4, to enhance patient education regarding cancer clinical trials. By leveraging informed consent forms from ClinicalTrials.gov, the researchers evaluated 2 artificial intelligence (AI)-driven approaches-direct summarization and sequential summarization-to generate patient-friendly summaries."
   Results: Multiple-Choice Question-Answer pairs "showed high concordance with human-annotated responses, and over 80% of surveyed participants reported enhanced understanding of the author's in-house BROADBAND trial. While LLMs hold promise in transforming patient engagement through improved accessibility of clinical trial information, concerns regarding AI hallucinations, accuracy, and ethical considerations remain." Models that existed 3 months prior to the 3/18/25 publication date include the 12/5/24 OpenAI o1 and 10/22/24 Claude 3.5 Sonnet. Utilization of these two models would have reduced broad speculation of LLM applications for future clinical trials.
2. **Title:** "Hallmarks of artificial intelligence contributions to precision oncology" 3/7/25 [20]
   Cons: Hallmark 1: cancer screening "AI has improved the efficiency and, in some cases, also the accuracy of screening compared to traditional methods." Hallmark 6: clinical trial design and matching "the recent advances in LLMs offer unprecedented opportunities for efficiently interpreting complex eligibility criteria within unstructured medical records.
   Results: The author's March 2025 10 Hallmarks provided a historical review to LLM precision oncology progression. More research articles are required in order to make rapid gains with LLM methods for projects at scale.

Conclusions: Large Scale LLM Oncology Trial Workflows Needed

1. Persistent publication of previously released LLMs is a result of an inefficient publication process.
2. End-to-end LLM workflows across trial phases for accelerating document and revision process.
3. LLM patient enrollment process is already an established large language model capability.
4. Excess number of review articles are less effective than clinical trial-type document iterations.
5. LLM workflow projections should decrease, while large repository scale LLM projects prosper.

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